

power consumption, and reduced thermal specs, there are also a number of new features inherent in DDR2. These are basically refinements that make DDR2 better, sort of an evolution over DDR1. Let's take a look at some of these.

1. ODT (On Die Termination): Perhaps one of the major drawbacks to extracting more speed (MHz) out of memory. Any signal moving along a bus reflects to a certain degree when it hits its intended target. These signal directions could go either way along the bus, and the reflected signal (or mirror signal) causes interference in the original signal, or could even cancel out the original signal depending on the original signal strength. This is where ODT scores, big time! DDR2 simply introduces a termination point to the original signal (DDR doesn't have ODT—refer to the features table above) once it has reached its target, by adding a resistor to ground it. This eliminates any reflection voltage as the resistor simply swallows up any signal voltage coming its way.

2. Posted CAS: Another necessary feature that DDR2 implements, whose basic aim is to eliminate data collisions along the bus. This is especially significant for DDR2 because of the high clock speeds involved, which increase the chances of collision. The command bus is responsible for issuing commands. A command buffer is placed on the DRAM chip that “bundles” these commands, that is, holds a command and schedules it for a later release. Therefore the command is pre-issued and stored, but the read operation is postponed. The command bus is freed from the burden of addressing exactly when to release that particular command. It can now activate the next bank. This delay is called Posted CAS. The delay that is specified during initialisation of the DRAM chip is called Additive Latency or AL. This, however, does play a part in increasing latency a bit; the read latency now becomes the sum total of the CAS latency and the Additive Latency. This isn't too detrimental to the bandwidth, however, because the command is issued early.

3. OCD (Off-Chip Driver) Calibration: DDR memory introduced the principle of clock forwarding. A single data strobe signal was used. The function of this strobe is to minimise signal skewness (“skewness” means deviations from the ideal curve that a signal is supposed to follow). Note that some amount of skewness is normal and cannot be avoided, but the goal

here is to minimise it, not eliminate it. This single strobe signal is compared to a reference point signal, which is predetermined. The problem of comparison becomes worse because the degree of skew is also never the same, and may change from clock to clock.

DDR2 introduces a bidirectional, differential strobe. This ensures higher signal integrity because it allows the two strobes to be calibrated against each other, as opposed to DDR, which simply calibrates on the basis of a reference point signal. With two reference points, skewness is minimised, and therefore even at higher frequencies, signal integrity is maintained.

Intel And AMD: Who Benefits More?

Looking at the comparative architectures of the Intel and AMD processors, it's quite easy to draw the conclusion that Intel CPUs are more memory bandwidth sensitive, rather than latency sensitive. In other words, an Intel processor performs better at higher DRAM frequencies, despite the higher latencies. In other words the Pentiums and their ilk are MHz demons, both clock-wise and memory requirement-wise!

The AMD 64 architecture is slightly different. AMD's 64-bit range has an integrated memory controller, which effectively negates the latencies that occur between the CPU and the Northbridge (where the MCH a.k.a. Memory Controller Hub typically resides). Due to this, AMD 64s become extremely latency sensitive, and such systems give blistering memory bandwidth scores. The controller can be as much as 90 per cent efficient when memory with tight timings is used. Compare this with Intel processors: their MCH isn't anywhere near as efficient. To make up for the efficiency of the controller, Intel processors need faster memory. They are perfect candidates to make use of the 800 MHz and 1066 MHz speeds that DDR2 dishes up.

This does not mean that AM2 processors are lacking in any way. In fact, the memory controller being efficient means that AM2s will be able to do wonders with the increased bandwidth. Although the latency will hit AMD harder, the silver lining is that DDR2 memory is continuously being refined, and memory with faster timings are being introduced every few months.

The drive for DDR2 is an obviously warranted one. The memory industry as a whole is still in a state of flux, with new technologies emerging and improvements being made. All this to feed the immense bandwidth requirements of today's users. Yes, that's correct, it's we users and our applications that drive the hardware industry as a whole, and not the other way round! The fact that DDR2 might well be a stop-gap for three to four years should also not escape you, with new technologies like DDR3 promising an emergence into the mainstream memory market. Remember when people said nothing could beat Dual DDR 400? Well, it just got beat—all the better for us! ■

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DDR And DDR2 At A Glance		
SPECIFICATIONS	DDR	DDR2
Clock speeds	200/266/333/400/500	400/533/667/800/1066
I/O Buffer frequency	100/133/166/200/250	200/266/333/400/533
DRAM core frequency	100/133/166/200/250	100/133/166/200/266
Prefetch size (in bits)	2	4
Voltage (volts)	2.5	1.8
Data strobe	Single DQS	Differential (DQS and /DQS)
Packaging	TSOP/TSOP 2	FBGA
Support for ODT	N	Y
OCD Calibration support	N	Y
Posted CAS	N	Y